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The Knee

A cross-sport comparison of performance-based outcomes of professional athletes following primary microfracture of the knee☆,☆☆

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ABSTRACT

Background: The purpose of this study was to compare performance-based outcomes among professional athletes in four major North American sports following microfracture to treat symptomatic chondral defects of the knee.
Methods: Major League Baseball (MLB), National Basketball Association (NBA), National Football League (NFL), and National Hockey League (NHL) athletes who underwent primary unilateral microfracture of the knee were identified through a previously reported protocol based on public sources. Successful return-to-play was defined as returning for at least one professional regular season game after surgery. Regular season player statistics and sport-specific performance scores were compiled for each player. Each player served as his own control, with the season prior to surgery defined as baseline. Comparisons across sports were enabled by adjusting for expected season and career length differences between sports and by calculating percent changes in performance.
Results: One hundred thirty one professional athletes who underwent microfracture were included. One hundred three athletes (78.6%) successfully returned to play. The ratio of games started-to-games played before surgery was found to be a significant positive independent predictor of returning ($p = 0.002$). Compared with their preoperative season, basketball and baseball players demonstrated significantly decreased performance one season after surgery (-14.8%, $p = 0.029$ and -12.9%, $p = 0.002$, respectively) that was recoverable to baseline by postoperative seasons 2–3 for baseball players but not for basketball players (-9.7%, $p = 0.024$).
Conclusion: Knee microfracture surgery is associated with a high rate of return to the professional level. However, the impact of this procedure on postoperative performance varied significantly depending on sport.

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1. Introduction

Professional athletes are at an increased risk of sustaining a variety of orthopedic injuries due to the considerable physical demands associated with elite levels of play. While the general population may be concerned with returning to activities of daily living, the physical demands of elite-level sports are much higher. Injuries of the knee, particularly damage to the articular

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cartilage, are common in the elite athlete population [1] and are potentially career-threating. As such, it is important for players, coaches, and team medical staff to be well aware of the potential career risks and benefits of surgical treatment for such injuries.

Articular cartilage injuries can be treated surgically with several different techniques including debridement/chondroplasty, marrow stimulation techniques (microfracture), osteochondral allograft/autograft transplantation, and autologous chondrocyte implantation [2]. Among these, microfracture, which involves the use of arthroscopic awls to make multiple holes into the subchondral bone beneath the chondral defect, is the most common [3]. This procedure produces local hemorrhage leading to the formation of a fibrin clot, which may promote differentiation and growth of chondral repair tissue into the defect [4,5].

Despite its frequent use, the durability of microfracture repairs has been questioned [6–8]. For example, in recreational athletes, microfracture surgery has been reported to lead to satisfactory short-term outcomes, but sub-optimal long-term results presumably from chronic cartilage degeneration. The professional athlete is exposed to far greater physical stress than a recreational competitor, warranting separate analyses and potential recommendations. In previous case series analyzing National Basketball Association (NBA) [9–11] and National Football League (NFL) [12–14] athletes, return-to-play (RTP) rates have been analyzed. However, as it currently stands the effect that knee microfracture has on return rates for Major League Baseball (MLB) and National Hockey League (NHL) athletes and on athletic performance compared with pre-injury levels for professional athletes in these four different sports has not been characterized. In addition, there is no consensus on whether certain sports portend worse outcomes following microfracture. It is reasonable to surmise that the varied physical demands from each professional sport may lead to different outcomes.

The purpose of this study was to perform a comparative assessment of performance-based outcomes among professional athletes in four major North American sports following microfracture to treat symptomatic chondral defects of the knee. We hypothesized professional athletes undergoing this procedure would be able to return to their previous level of sport and performance at a high rate after surgery.

2. Methods

2.1. Inclusion criteria

A well-established algorithmic protocol [15–20] based on public sources was utilized to create a database of MLB, NBA, NFL (American football, hereafter referred to as “football”), and NHL athletes who underwent microfracture of the knee. NewsBank® search engine was used to identify players from each sport who underwent surgery, using the search terms “[professional league] knee microfracture surgery”. Player name and date of surgery were recorded for players captured by this initial query. Next, each case was validated individually using player profiles, newspaper archives, press releases, and team injury reports using the more specific search terms “[player name] knee microfracture surgery”. Inclusion criteria for this study were MLB, NBA, NFL, and NHL athletes who reportedly underwent primary unilateral microfracture surgery to treat symptomatic chondral defects of the knee while on the active roster of a professional team and had a definitive date of surgery. For each case, two independent sources of record were cross-referenced to confirm that a microfracture procedure was performed as defined by the inclusion criteria. Newly drafted players who underwent microfracture prior to playing in any professional regular season games or rookie players who underwent season-ending or non-season-ending surgery were included only for RTP analysis. Players who reportedly underwent multiple microfracture procedures (revision and/or bilateral), had the procedure while in college or while on a minor league team, retired before surgery, did not yet have postoperative data available (e.g., underwent surgery too recently), or had conflicting reports from different sources were excluded.

2.2. Performance-based outcome measures

Demographic data was manually collected for each athlete, including date of birth, weight, height, date of professional regular season debut, position, date of most recent professional regular season game, and retirement date (when applicable). Successful RTP was defined as returning for at least one regular season game in a major professional league after surgery, excluding minor leagues. This definition was chosen in order to standardize comparisons and minimize potential confounders, given the greater physical demands and skill level required for major league-level play compared with other levels (e.g., minor or international leagues). Further, professional game participation data can be readily collected from public sources, whereas player data at other possible return points (e.g., first postoperative workout, practice, or scrimmage) is not consistently reported. Return time was defined as the number of days elapsed between reported date of surgery and date of first professional game played. The index season was defined as the professional regular season immediately before surgery (off-season surgeries) or the prior season (season-ending and non-season-ending surgeries). For season-ending and off-season surgeries, the season corresponding to a player's first professional game back was defined as postoperative season one. For non-season-ending surgeries, the professional season after the season in which surgery occurred was defined as postoperative season one. Regular season game statistics were compiled both pre- and post-surgery, including all-star appearances, games played, games started (except hockey, in which players rotate for each position), and performance score (PS). Two statistical time points were used after microfracture: postoperative season one and an average of postoperative seasons two and three. Each player served as his own control, with performance compared to the index (baseline) season.

Performance scores were calculated for each sport for players who successfully returned using previously established and refined scoring systems. For baseball, performance scores were calculated using accepted sabermetrics based on a player's role as a hitter or

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pitcher. The number of walks plus hits per inning pitched (WHIP) was used as the PS for pitchers [21] and on-base plus slugging percentage (OPS) was used as the PS for hitters [22]. For basketball, player efficiency rating (PER) was used to quantify player performance [9,23]. Sport-specific scores were manually computed for football [16] and hockey [24] athletes. Rookies or first-year players and non-skill position football players (offensive linemen, long snappers, and punters) did not have statistics available to calculate PS and were excluded from performance evaluation. The ratio of games-started-to-games played (GS:GP) was calculated for each player as an additional measure of player performance [15].

Cross-sport comparisons were permitted by adjusting for expected season and career length differences between sports and by calculating percent changes in performance. Number of games played was normalized for each player to account for differences in typical regular season length between sports (MLB = 162 games; NBA = 82 games; NFL = 16 games; NHL = 82 games). Similarly, career lengths were standardized for each player by adjusting for differences in expected career length between sports (MLB = 5.6 years; NBA = 4.8 years; NFL = 3.5 years; NHL = 5.5 years) based on a previously published methodology [17]. This study was deemed exempt by the Institutional Review Board as all data was obtained from publicly available sources.

2.3. Statistical analysis

IBM SPSS Statistics software (version 24.0; IBM Inc., Armonk, New York, USA) was used for all statistical analyses. A Fisher exact test was performed to evaluate categorical variables. Continuous variables were evaluated using a two-tailed Student's *t*-Test for normally distributed data. Each player served as his own control, with changes in pre- and post-surgery statistics assessed using a two-tailed paired *t*-Test. Regression models were used to evaluate the effect of independent variables (age, experience, and body mass index [BMI]) on RTP and performance outcomes. Kaplan–Meier survivorship analysis was also performed with endpoint defined as official retirement from sport. For all statistical analyses, the threshold for significance was established at $p < 0.05$. Numerical values given were mean ± standard deviation unless noted otherwise and *p*-values correspond to each sport compared with all other sports combined.

3. Results

A total of 131 professional athletes (18 MLB, 34 NBA, 76 NFL, three NHL) who underwent microfracture between 1991 and 2015 were identified (Table 1). On average, athletes in the MLB cohort were significantly older at the time of surgery than all other athletes ($p = 0.001$) and NFL athletes were younger ($p = 0.007$). Normalized preoperative career experience (total years and games played) was comparable between sports, with no statistically significant differences.

A total of 103 of 131 athletes (78.6%) treated with microfracture successfully returned to play, which varied according to sport (Figure 1). MLB athletes demonstrated a significantly higher return rate than all other sports (100.0% vs. 75.0%, respectively; $p = 0.013$), while NFL athletes had a lower rate of return (71.1% vs. 89.0%, respectively; $p = 0.017$). Average return time was 293 ± 146 days, which was significantly longer for NFL athletes compared with all other sports (327 ± 143 vs. 255 ± 142 days, respectively, $p = 0.013$) (Table 2).

Kaplan–Meier survivorship analysis demonstrated that 67% and 43% of professional athletes who successfully returned were expected to still be on active rosters in their respective leagues at the one- and three-year time points after surgery, respectively, with no significant differences between sports (Figure 2). Overall median (50%) survival was 2.8 years after surgery, with the highest occurring among MLB athletes (4.5 years) and lowest for NFL athletes (2.1 years).

The percentage of games started versus played (GS:GP) before surgery was significantly higher among athletes who successfully returned compared with those who did not during players' index seasons (0.75 vs. 0.52, $p = 0.005$) and total preoperative careers (0.71 vs. 0.48, $p = 0.001$). Binomial logistic regression confirmed that total preoperative career GS:GP was a positive independent

Table 1
Summary of average age, body mass index (BMI), and pre-surgery career statistics for professional athletes included in this study, organized by sport.

	MLB (n = 18)	NBA (n = 34)	NFL (n = 76)	NHL (n = 3)
Age (years)	31.1 ± 3.5[a]	28.1 ± 4.3	27.4 ± 3.3[a]	32.0 ± 4.7
All other players (years)	27.8 ± 3.7	28.3 ± 3.7	29.3 ± 4.3	28.1 ± 3.8
p-Value	**0.001**	0.857	**0.007**	0.292
BMI (kg/m^2)	28.1 ± 1.7[a]	25.6 ± 1.7[a]	32.5 ± 4.3[a]	26.7 ± 0.23[a]
All other players (years)	30.3 ± 4.9	31.5 ± 4.4	26.5 ± 2.0	30.0 ± 4.6
p-Value	**0.001**	<**0.001**	<**0.001**	<**0.001**
Career before surgery[b] (years)	7.3 ± 3.6	7.4 ± 4.9	7.1 ± 5.4	12.2 ± 6.2
All other players (years)	7.3 ± 5.3	7.3 ± 5.2	7.6 ± 4.6	7.2 ± 5.0
p-Value	0.974	0.927	0.537	0.296
Games played before surgery[b]	679 ± 498	860 ± 635	659 ± 515	1537 ± 1024
All other players (years)	743 ± 584	690 ± 544	838 ± 632	715 ± 550
p-Value	0.630	0.171	0.088	0.299

Significance was defined as $p \leq 0.05$.
[a] Denotes significance compared with all other players.
[b] Reflects adjusted values to account for differences in average expected career length and regular season length between sports.

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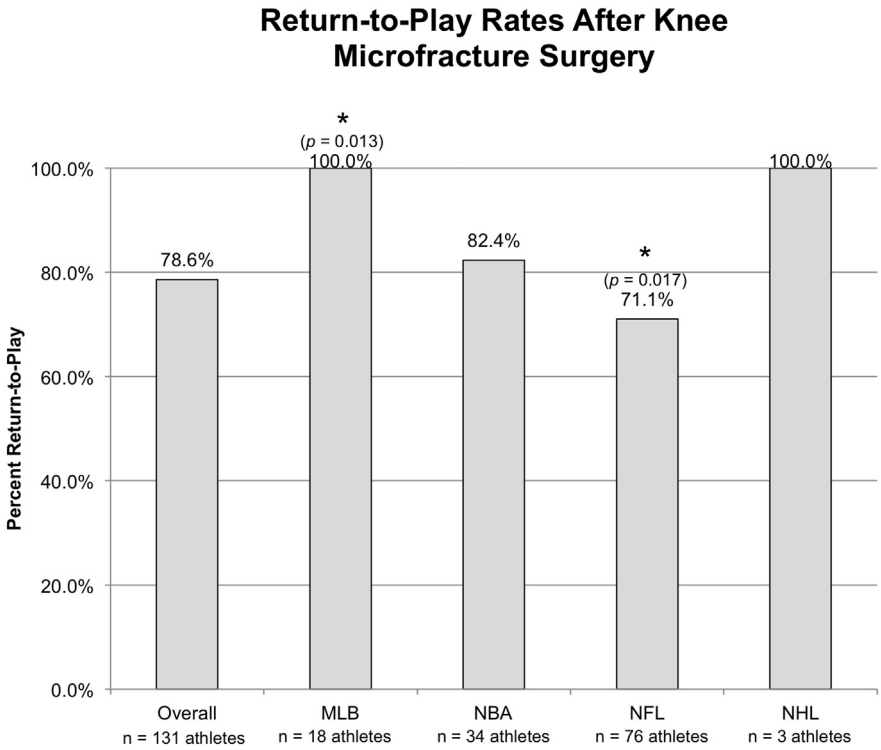

Figure 1. Return-to-play rates following knee microfracture surgery for American professional athletes. *Denotes significance compared with all other sports combined.

predictor of returning ($p = 0.002$). Other demographic variables including age, BMI, and experience before surgery (years played, total games played, and games played during index season) were not significantly different between athletes who returned and those who did not.

Of the 103 players who returned, 82 (80%) met the inclusion criteria for PS evaluation (18 MLB, 28 NBA, 36 NFL) and had at least one-year follow-up data available, with 40 (39%; 10 MLB, 15 NBA, 15 NFL) having two or more years of follow-up data available. There were 13 NFL offensive linemen and two NFL rookies who did not have statistics available to calculate PS and were excluded from performance evaluation. The small number of NHL athletes (three) was not amenable to statistical analysis of performance variables and these players were excluded from performance evaluation. Finally, three NFL athletes underwent non-season-ending surgery and retired after returning (before postoperative season one as defined by this study). Differences in change in performance (career before vs. career after procedure), adjusted career length, and adjusted games played after surgery between individual sports were not significant.

NBA athletes who successfully returned to play showed a significant reduction in adjusted games played during their first season post-surgery compared to baseline (61 vs. 100, respectively, $p < 0.0001$). NBA and MLB athletes demonstrated significantly decreased performance compared to the index season during postoperative season one ($p = 0.029$ and $p = 0.002$, respectively). NBA athletes also demonstrated significantly decreased performance during postoperative seasons two to three ($p < 0.024$) (Figure 3), an effect not seen in any other sport. Notably, NFL athletes, who comprised the greatest percentage players in this study, did not demonstrate any reductions in statistical performance after return to play.

Table 2
Performance outcomes for professional athletes who underwent microfracture, organized by sport.

League	Return (rate)	RTP time (days)	Career after surgery (years)[d]	Games played after surgery[d]
MLB	18/18 (100.0%)[a]	254 ± 133	3.2 ± 3.5	311 ± 328
NBA	28/34 (82.4%)	255 ± 156	4.1 ± 3.9	420 ± 446
NFL	54/76 (71.1%)[b]	327 ± 143[c]	3.6 ± 4.0	383 ± 342
NHL	3/3 (100.0%)	258 ± 72	2.9 ± 2.6	352 ± 314

[a] Denotes significance compared with all other players (100.0% vs. 75.2%, $p = 0.013$).
[b] Denotes significance compared with all other players (71.1% vs. 89.1%, $p = 0.017$).
[c] Denotes significance compared with all other players (327 ± 143 vs. 255 ± 142, $p = 0.013$).
[d] Reflects adjusted values to account for differences in average expected career length and regular season length between sports.

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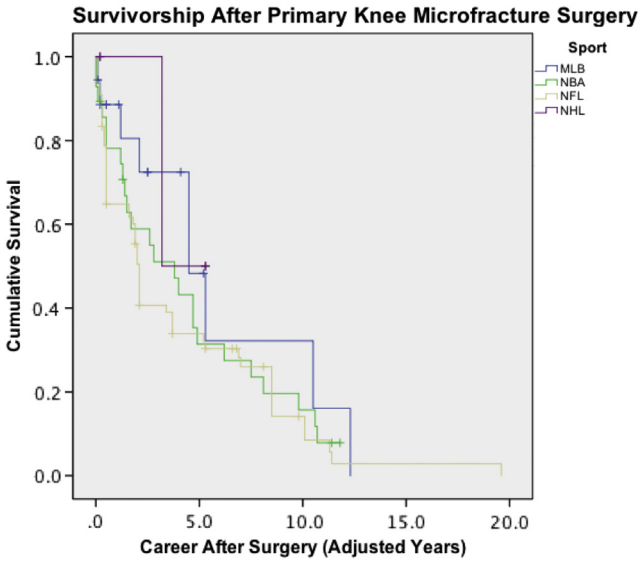

Figure 2. Kaplan–Meier survivorship analysis after knee microfracture surgery for American professional athletes, with survivorship defined as official retirement from sport. Highest median survival occurred among MLB players (4.5 years) and lowest median survival occurred among NFL players (2.1 years). No significant differences between sports were identified.

4. Discussion

The data from this study support part of our hypothesis, in that microfracture is associated with a high rate of return to the professional level, with 78.6% of players able to successfully return, which is in concordance with other studies [9–14]. Notably, the proportion of games started during an athlete's entire preoperative career was found to be a significant positive independent predictor of returning after surgery. This finding may be at least partly explained by the financial implications associated with returning, as starting players likely have a greater financial incentive to return compared with non-starters.

Contrary to our hypothesis, performance-based outcomes following microfracture varied significantly depending on sport. For example, while all MLB players were able to successfully return to professional play and returned to baseline performance within three seasons of surgery, outcomes were significantly different for NBA and NFL players. NFL players returned at a significantly lower rate and required a longer recovery time than players in the three other sports. However, while these athletes had a more challenging time returning to play, those that did were able to quickly return to their preoperative level of performance. Conversely, NBA players returned at a higher rate than NFL players, but short- and long-term postoperative performance

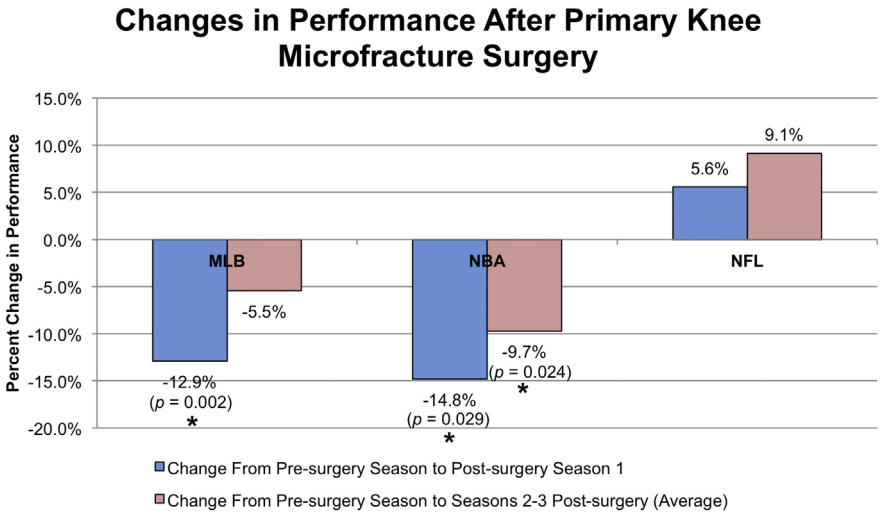

Figure 3. Performance-based outcomes among American professional athletes following knee microfracture surgery. *Denotes significance compared to index season. NHL excluded due to small sample size.

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compared to baseline was significantly lower among NBA players. These findings suggest that the relative physical demands for each sport may have a direct impact on return rates after knee microfracture. For example, in baseball, with considerably less running, jumping, and direct impact than the other sports, return rates can be expected to be high. Conversely, football and basketball involve repetitive activities that place a great deal of stress on the knee (e.g., jumping, cutting) and players are prone to injuries that can directly damage knee cartilage (e.g., falling from height on a hardwood court in basketball, being tackled or cut-blocked in football). These risk factors may predispose football and basketball players to decreased return rates, slower return times, and/or decreased long-term performance. Other factors, including position, duration of symptoms, location of the microfracture, concomitant procedures, and quality of rehabilitation may also explain our findings, as these have a large role in whether a player is able to return and his level of performance. Accordingly, it is our recommendation that type of sport, position, extent of chondral injury and other concomitant injuries, and preoperative level of function (including performance) be considered when evaluating a player's suitability for microfracture and when counseling players on postoperative expectations.

Psychosocial factors may also help to explain the observed differences in return rates and postoperative performance in these athletes. First, roster changes, coaching decisions, and individual contract-specific issues may all have an impact on a player's ability to return and his level of participation after surgery. Second, with only five players on an NBA court versus 11 for an NFL field, there may be differences in the exposure of a player's limitations after knee microfracture, which may lead to variable playing opportunities. The realities of contract implications also cannot be ignored. There may be more incentive for the NFL athlete to return at a high level immediately after surgery since player contracts are not guaranteed while for NBA players, the guaranteed contract may lead to a predisposition for a more gradual return to baseline performance [25]. Finally, there are also differences in the structure of the professional leagues included in this study, such as all leagues except the NFL having a “minor league” that may allow players a better chance of returning to an elite level after surgery.

Interestingly, the findings from this study directly contrast with outcomes after anterior cruciate ligament reconstruction (ACLR) in this patient population. Previous studies have concluded that NFL athletes demonstrated significantly reduced performance following ACLR up to three seasons postoperatively [12], while NBA players recover to baseline after two years [25]. The comparison of these outcomes with those after microfracture highlights the potential impact of sport on postoperative expectations after different procedures of the knee. The daily collision of American football players may place greater stress on an anterior cruciate ligament (ACL)-reconstructed knee compared to one with microfracture. The demands of running on grass/turf may lead to a lower risk for symptoms of an osteochondral defect. Similarly, a basketball player's ability to regain quickness and vertical jump after ligament reconstruction may be more successful than one with a cartilage injury. These findings may provide new insight into the implications of cartilaginous versus ligamentous injuries of the knee and may highlight heterogeneity in postoperative rehabilitation protocols across sports.

4.1. Limitations

This study has several limitations that have been reported and discussed in previous retrospective studies [12,15–17]. First, the use of public sources to identify players who underwent knee microfracture and no other potentially confounding procedures has inherent limitations. Second, the capture rate of players who underwent this procedure comes into question when using public sources. For example, individual policies and preferences of leagues, teams, and players may have introduced sampling bias towards more high profile players. However, we believe with the widespread media attention given to and the increased understanding of the layperson regarding this procedure that search terms using “microfracture” with our methodology would identify players who underwent this operation. Third, our methodology using public sources represents the only way to evaluate surgical outcomes for a large group of professional athletes across multiple sports, as single-surgeon case series often are limited to one sport and/or smaller sample sizes. However, with our methodology there was the potential for errors and omissions in the available data, which may have influenced the results. To this end, the study period was limited to a relatively recent time period, such that more reliable, redundant reporting was available. Each case was extensively screened using multiple sources in order to confirm that knee microfracture was performed and any cases that had conflicting or ambiguous reports were excluded. Even though microfracture was reported as the primary operation for all cases in this study, it is still possible that some players had additional concomitant procedures performed during knee arthroscopy (e.g., chondroplasty, meniscal repair, augmentation of microfracture with biologics) that may have influenced the results. Fourth, specific information regarding diagnosis (e.g., presenting symptoms, etiology, severity, diagnostic imaging, alignment, or size, number, severity, and location of chondral lesions), chronicity of symptoms, previous conservative or surgical treatments, other concomitant pathology, specific surgical indications, intraoperative details, postoperative management and rehabilitation compliance, follow-up examinations (e.g., imaging, assessment of readiness/clearance for return), and complications was not available with our methodology. Finally, this study only evaluated males, which may not necessarily apply to female professional athletes.

5. Conclusions

The results of this study support the hypothesis that knee microfracture surgery is associated with a high rate of return to the professional level in four major North American sports organizations. However, the impact of this procedure on postoperative performance varied significantly depending on sport. Further research is warranted comparing return rates and performance outcomes for microfracture with different cartilage repair procedures (e.g., autologous chondrocyte implantation) in the elite athlete population.

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References

[1] DeHaven KE, Lintner DM. Athletic injuries: comparison by age, sport, and gender. Am J Sports Med 1986;14:218–24.
[2] Rodríguez-Merchán EC, De la Corte-García H. Regeneration of articular cartilage of the knee: basic concepts. In: Rodrìguez-Merchán CE, editor. Articular cartilage defects of the knee: diagnosis and treatment. Milano: Springer Milan; 2012. p. 1–16.
[3] Mithoefer K, Cole BJ, Sekiya JK. Microfracture technique in the knee. Surgical techniques of the shoulder, elbow, and knee in sports medicine. 2 ed. Philadelphia, PA: Elsevier Saunders; 2013; 663–70.
[4] Oussedik S, Tsitskaris K, Parker D. Treatment of articular cartilage lesions of the knee by microfracture or autologous chondrocyte implantation: a systematic review. Arthroscopy 2015;31:732–44.
[5] Salzmann GM, Sah B, Südkamp NP, Niemeyer P. Clinical outcome following the first-line, single lesion microfracture at the knee joint. Arch Orthop Trauma Surg 2013;133:303–10.
[6] Gobbi A, Karnatzikos G, Kumar A. Long-term results after microfracture treatment for full-thickness knee chondral lesions in athletes. Knee Surg Sports Traumatol Arthrosc 2014;22:1986–96.
[7] Kreuz PC, Steinwachs MR, Erggelet C, Krause SJ, Konrad G, Uhl M, et al. Results after microfracture of full-thickness chondral defects in different compartments in the knee. Osteoarthritis Cartilage 2006;14:1119–25.
[8] Mithoefer K, Williams III RJ, Warren RF, Wickiewicz TL, Marx RG. High-impact athletics after knee articular cartilage repair: a prospective evaluation of the microfracture technique. Am J Sports Med 2006;34:1413–8.
[9] Cerynik DL, Lewullis GE, Joves BC, Palmer MP, Tom JA. Outcomes of microfracture in professional basketball players. Knee Surg Sports Traumatol Arthrosc 2009; 17:1135–9.
[10] Harris JD, Walton DM, Erickson BJ, Verma NN, Abrams GD, Bush-Joseph CA, et al. Return to sport and performance after microfracture in the knees of National Basketball Association players. Orthop J Sports Med 2013;1:2325967113512759.
[11] Minhas SV, Kester BS, Larkin KE, Hsu WK. The effect of an orthopaedic surgical procedure in the National Basketball Association. Am J Sports Med 2016;44: 1056–61.
[12] Mai HT, Alvarez AP, Freshman RD, Chun DS, Minhas SV, Patel AA, et al. The NFL Orthopaedic Surgery Outcomes Database (NO-SOD): the effect of common orthopaedic procedures on football careers. Am J Sports Med 2016;44(9):2257.
[13] Scillia AJ, Aune KT, Andrachuk JS, Cain EL, Dugas JR, Fleisig GS, et al. Return to play after chondroplasty of the knee in National Football League athletes. Am J Sports Med 2015;43:663–8.
[14] Steadman JR, Miller BS, Karas SG, Schlegel TF, Briggs KK, Hawkins RJ. The microfracture technique in the treatment of full-thickness chondral lesions of the knee in National Football League players. J Knee Surg 2003;16:83–6.
[15] Hsu WK. Performance-based outcomes following lumbar discectomy in professional athletes in the National Football League. Spine 2010;35:1247–51.
[16] Hsu WK. Outcomes following nonoperative and operative treatment for cervical disc herniations in National Football League athletes. Spine 2011;36:800–5.
[17] Hsu WK, McCarthy KJ, Savage JW, Roberts DW, Roc GC, Micev AJ, et al. The Professional Athlete Spine Initiative: outcomes after lumbar disc herniation in 342 elite professional athletes. Spine J 2011;11:180–6.
[18] Parekh SG, Wray III WH, Brimmo O, Sennett BJ, Wapner KL. Epidemiology and outcomes of Achilles tendon ruptures in the National Football League. Foot Ankle Spec 2009;2(6):283.
[19] Roberts DW, Roc GJ, Hsu WK. Outcomes of cervical and lumbar disk herniations in Major League Baseball pitchers. Orthopedics 2011;34:602–9.
[20] Savage JW, Hsu WK. Statistical performance in National Football League athletes after lumbar discectomy. Clin J Sport Med 2010;20:350–4.
[21] Fedoriw WW, Ramkumar P, McCulloch PC, Lintner DM. Return to play after treatment of superior labral tears in professional baseball players. Am J Sports Med 2014;42:1155–60.
[22] Laby DM, Kirschen DG, De Land P. The effect of laser refractive surgery on the on-field performance of professional baseball players. Optometry 2005;76:647–52.
[23] Busfield BT, Kharrazi FD, Starkey C, Lombardo SJ, Seegmiller J. Performance outcomes of anterior cruciate ligament reconstruction in the National Basketball Association. Arthroscopy 2009;25:825–30.
[24] Schroeder GD, McCarthy KJ, Micev AJ, Terry MA, Hsu WK. Performance-based outcomes after nonoperative treatment, discectomy, and/or fusion for a lumbar disc herniation in National Hockey League athletes. Am J Sports Med 2013;41:2604–8.
[25] Kester BS, Behery OA, Minhas SV, Hsu WK. Athletic performance and career longevity following anterior cruciate ligament reconstruction in the National Basketball Association. Knee Surg Sports Traumatol Arthrosc 2016:1–7.

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